

Beyond Ingress

Mastering the Kubernetes Gateway API

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Agenda



Expose Pods

Gateway API

Routing

Gateway API vs. Ingress

Gateway Controller

Demo

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Expose Pods

Expose Pods

Service with type Load Balancer

Ingress

Gateway API

Ingress and Gateway API Features

- One entry point
- Better routing
- Automate TLS certificates with Cert-Manager
- TLS termination

Ingress

Ingress Controller:

- Traefik
- Nginx
- Application Gateway Ingress Controller (AGIC)

Ingress Downsides

Ingress controllers have a lot of control how traffic is routed

Annotations are unique to the used Ingress controller (e.g. Nginx) → vendor lock-in

No cross-namespace support for TLS secrets

Ingress Downsides

Ingress does not support traffic policies

No modularity

No real Separation of Concerns

Gateway API announced in 2023 to solve these downside

Gateway API

Gateway API

Components are modular and split into multiple parts

Gateway Class configures proxy (similar to Ingress Class)

Gateway Class depends on Gateway Controller

```
apiVersion: gateway.networking.k8s.io/v1
kind: GatewayClass ←
metadata:
  name: envoy
spec:
  controllerName: gateway.envoyproxy.io/gatewayclass-controller
```

```
apiVersion: gateway.networking.k8s.io/v1
kind: GatewayClass
metadata:
  name: envoy ←
spec:
  controllerName: gateway.envoyproxy.io/gatewayclass-controller
```

```
apiVersion: gateway.networking.k8s.io/v1
```

```
kind: GatewayClass
```

```
metadata:
```

```
  name: envoy
```

```
spec:
```

```
  controllerName: gateway.envoyproxy.io/gatewayclass-controller
```

Gateway

Gateway contains listeners

Governs TLS certificates

Namespaced

Gateway consumes global Gateway Class

apiVersion: gateway.networking.k8s.io/v1

kind: Gateway 

metadata:

name: Gateway

namespace: gateway-ns

annotations:

cert-manager.io/cluster-issuer: clusterissuer

spec:

gatewayClassName: envoy

listeners:

- name: http-listener

port: 80

protocol: HTTP

allowedRoutes:

namespaces:

from: All

apiVersion: gateway.networking.k8s.io/v1

kind: Gateway

metadata:

name: Gateway

namespace: gateway-ns

annotations:

cert-manager.io/cluster-issuer: clusterissuer

spec:

gatewayClassName: envoy

listeners:

- name: http-listener

port: 80

protocol: HTTP

allowedRoutes:

namespaces:

from: All

```
apiVersion: gateway.networking.k8s.io/v1
```

```
kind: Gateway
```

```
metadata:
```

```
  name: Gateway
```

```
  namespace: gateway-ns
```

```
  annotations:
```

```
    cert-manager.io/cluster-issuer: clusterissuer ←
```

```
spec:
```

```
  gatewayClassName: envoy
```

```
  listeners:
```

```
  - name: http-listener
```

```
    port: 80
```

```
    protocol: HTTP
```

```
    allowedRoutes:
```

```
      namespaces:
```

```
        from: All
```

apiVersion: gateway.networking.k8s.io/v1

kind: Gateway

metadata:

name: Gateway

namespace: gateway-ns

annotations:

cert-manager.io/cluster-issuer: clusterissuer

spec:

gatewayClassName: envoy ←

listeners:

- name: http-listener

port: 80

protocol: HTTP

allowedRoutes:

namespaces:

from: All

apiVersion: gateway.networking.k8s.io/v1

kind: Gateway

metadata:

name: Gateway

namespace: gateway-ns

annotations:

cert-manager.io/cluster-issuer: clusterissuer

spec:

gatewayClassName: envoy

listeners:

- name: http-listener

port: 80

protocol: HTTP

allowedRoutes:

namespaces:

from: All

- name: traefik-https-listener

port: 443

protocol: HTTPS

hostname: demo.programmingwithwolfgang.com

allowedRoutes:

namespaces:

from: All

tls:

certificateRefs:

- group: ""

kind: Secret

name: demo-secret

```
- name: traefik-https-listener
  port: 443
  protocol: HTTPS
  hostname: demo.programmingwithwolfgang.com
  allowedRoutes:
    namespaces:
      from: All
  tls:
    certificateRefs:
      - group: ""
        kind: Secret
        name: demo-secret
```



```
- name: traefik-https-listener
  port: 443
  protocol: HTTPS
  hostname: demo.programmingwithwolfgang.com
  allowedRoutes:
    namespaces:
      from: All
  tls:
    certificateRefs:
      - group: ""
        kind: Secret
        name: demo-secret
```

ListenerSet

Allow teams to configure their listeners

Automate TLS certificate management

Perfect for multitenancy

Reduces the blast radius

Gateways can only have 64 listeners


```
apiVersion: gateway.networking.k8s.io/v1
```

```
kind: ListenerSet
```

```
metadata:
```

```
  name: demo-listenerset
```

```
  namespace: demo-app-ns
```

```
spec:
```

```
  parentRef:
```

```
    group: gateway.networking.k8s.io
```

```
    kind: Gateway
```

```
    name: Gateway
```

```
    namespace: gateway-ns
```

```
  listeners:
```

```
    - name: http-listener
```

```
      protocol: HTTP
```

```
      port: 80
```

```
      hostname: myapp.programmingwithwolfgang.com
```

```
apiVersion: gateway.networking.k8s.io/v1
```

```
kind: ListenerSet
```

```
metadata:
```

```
  name: demo-listenerset
```

```
  namespace: demo-app-ns
```

```
spec:
```

```
  parentRef:
```

```
    group: gateway.networking.k8s.io
```

```
    kind: Gateway
```

```
    name: Gateway
```

```
    namespace: gateway-ns
```

```
  listeners:
```

```
    - name: http-listener
```

```
      protocol: HTTP
```

```
      port: 80
```

```
      hostname: myapp.programmingwithwolfgang.com
```

```
apiVersion: gateway.networking.k8s.io/v1
```

```
kind: ListenerSet
```

```
metadata:
```

```
  name: demo-listenerset
```

```
  namespace: demo-app-ns
```

```
spec:
```

```
  parentRef:
```

```
    group: gateway.networking.k8s.io
```

```
    kind: Gateway
```

```
    name: Gateway
```

```
    namespace: gateway-ns
```

```
  listeners:
```

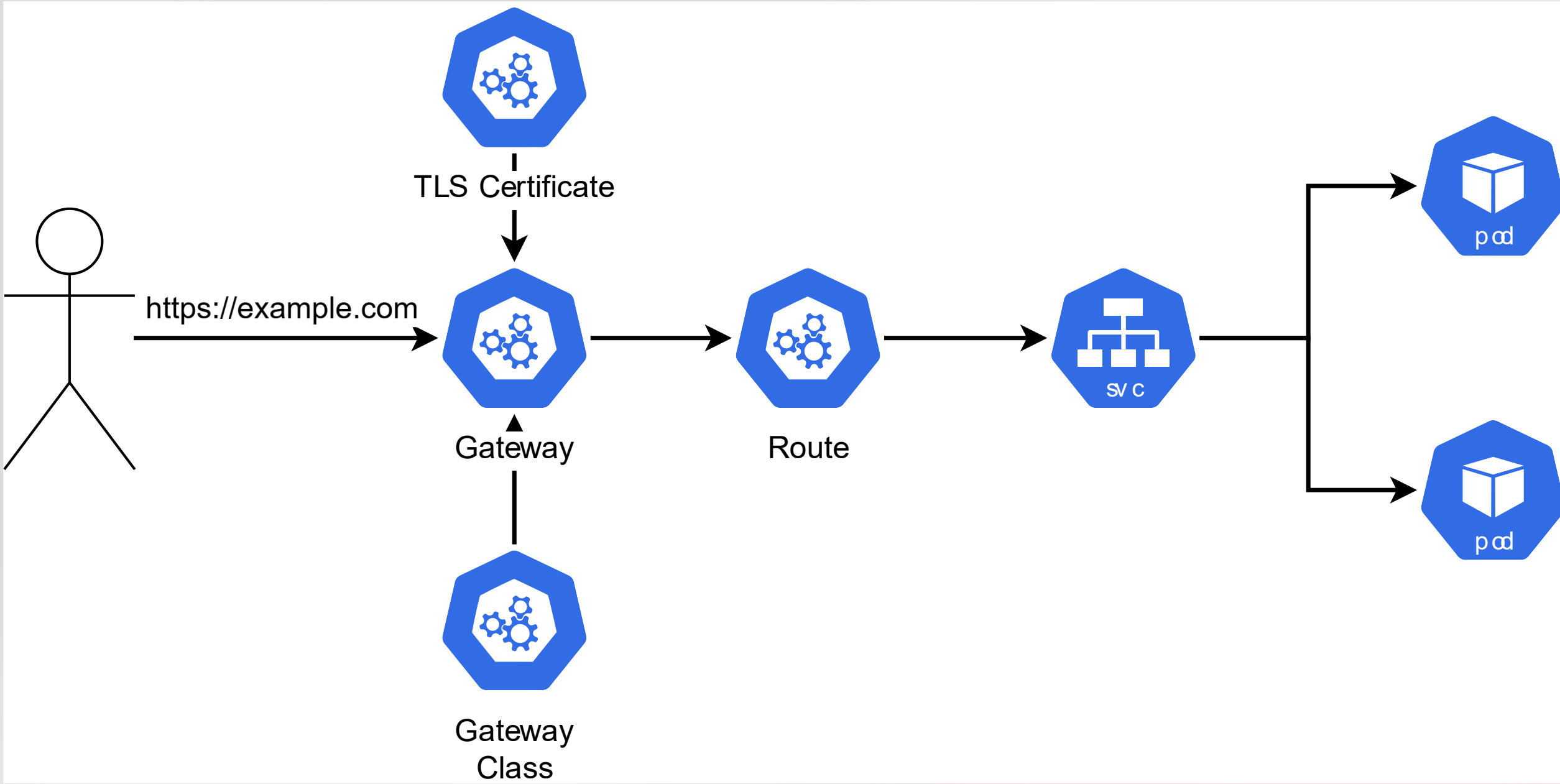
```
    - name: http-listener
```

```
      protocol: HTTP
```

```
      port: 80
```

```
      hostname: myapp.programmingwithwolfgang.com
```

```
apiVersion: gateway.networking.k8s.io/v1
kind: ListenerSet
metadata:
  name: demo-listenerset
  namespace: demo-app-ns
spec:
  parentRef:
    group: gateway.networking.k8s.io
    kind: Gateway
    name: Gateway
    namespace: gateway-ns
  listeners:
    - name: http-listener
      protocol: HTTP
      port: 80
      hostname: myapp.programmingwithwolfgang.com
```



Routing

Routing

Listeners can listen on different ports

Different protocols:

- HTTP
- HTTPS
- TLS
- TCP
- UDP

Routes

Uses standards instead of annotations

Routes:

- HTTPS and HTTP
- TCP and UDP
- GRPC
- TLS


```
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute ←
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
        backendRefs:
          - name: traefik
            port: 80
```

```
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
        backendRefs:
          - name: traefik
            port: 80
```

```
apiVersion: gateway.networking.k8s.io/v1
```

```
kind: HTTPRoute
```

```
metadata:
```

```
  name: traefik-http-route
```

```
  namespace: traefik-ns
```

```
spec:
```

```
  parentRefs:
```

```
    - name: gateway
```

```
      namespace: gateway-ns
```

```
  hostnames:
```

```
    - "demo.programmingwithwolfgang.com" ←
```

```
  rules:
```

```
    - matches:
```

```
      backendRefs:
```

```
        - name: traefik
```

```
          port: 80
```

```
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
        backendRefs:
          - name: traefik
            port: 80
```

Routing

Routing options:

- Path
- Header, e.g. User-Agent
- Query
- HTTP method
- Redirect request

hostnames:

- "demo.programmingwithwolfgang.com"

rules:

- backendRefs:

- name: app-one ←

port: 80

weight: 20 ←

- name: app-two

port: 80

weight: 80

hostnames:

- "demo.programmingwithwolfgang.com"

rules:

- backendRefs:

- name: app-one

port: 80

weight: 20

- name: app-two ←

port: 80

weight: 80 ←

rules:

- matches:
 - path:
 - type: PathPrefix
 - value: /my/path
- filters:
 - type: URLRewrite
 - urlRewrite:
 - path:
 - type: ReplaceFullPath
 - replaceFullPath: /replaced
- backendRefs:
 - name: app-one
 - port: 80
- backendRefs:
 - name: app-two
 - port: 80

rules:

- matches:
 - path:
 - type: PathPrefix
 - value: /my/path

filters:

- type: URLRewrite
 - urlRewrite:
 - path:
 - type: ReplaceFullPath
 - replaceFullPath: /replaced

backendRefs:

- name: app-one
 - port: 80
- backendRefs:
 - name: app-two
 - port: 80

rules:

- matches:
 - path:
 - type: PathPrefix
 - value: /my/path
 - filters:
 - type: URLRewrite
 - urlRewrite:
 - path:
 - type: ReplaceFullPath
 - replaceFullPath: /replaced

backendRefs:

- name: app-one
 - port: 80
- backendRefs:
 - name: app-two
 - port: 80

rules:

- matches:
 - path:
 - type: PathPrefix
 - value: /my/path
 - filters:
 - type: URLRewrite
 - urlRewrite:
 - path:
 - type: ReplaceFullPath
 - replaceFullPath: /replaced
 - backendRefs:
 - name: app-one
 - port: 80
 - backendRefs:
 - name: app-two
 - port: 80

Gateway API vs Ingress

Gateway API vs Ingress

Gateway API might not replace Ingress

Ingress is very easy to manage

- Especially in smaller teams

Gateway might be better for multi-team setups due to modularity

Gateway Controller

Gateway Controller

Nginx Gateway Fabric

Traefik

Envoy Gateway

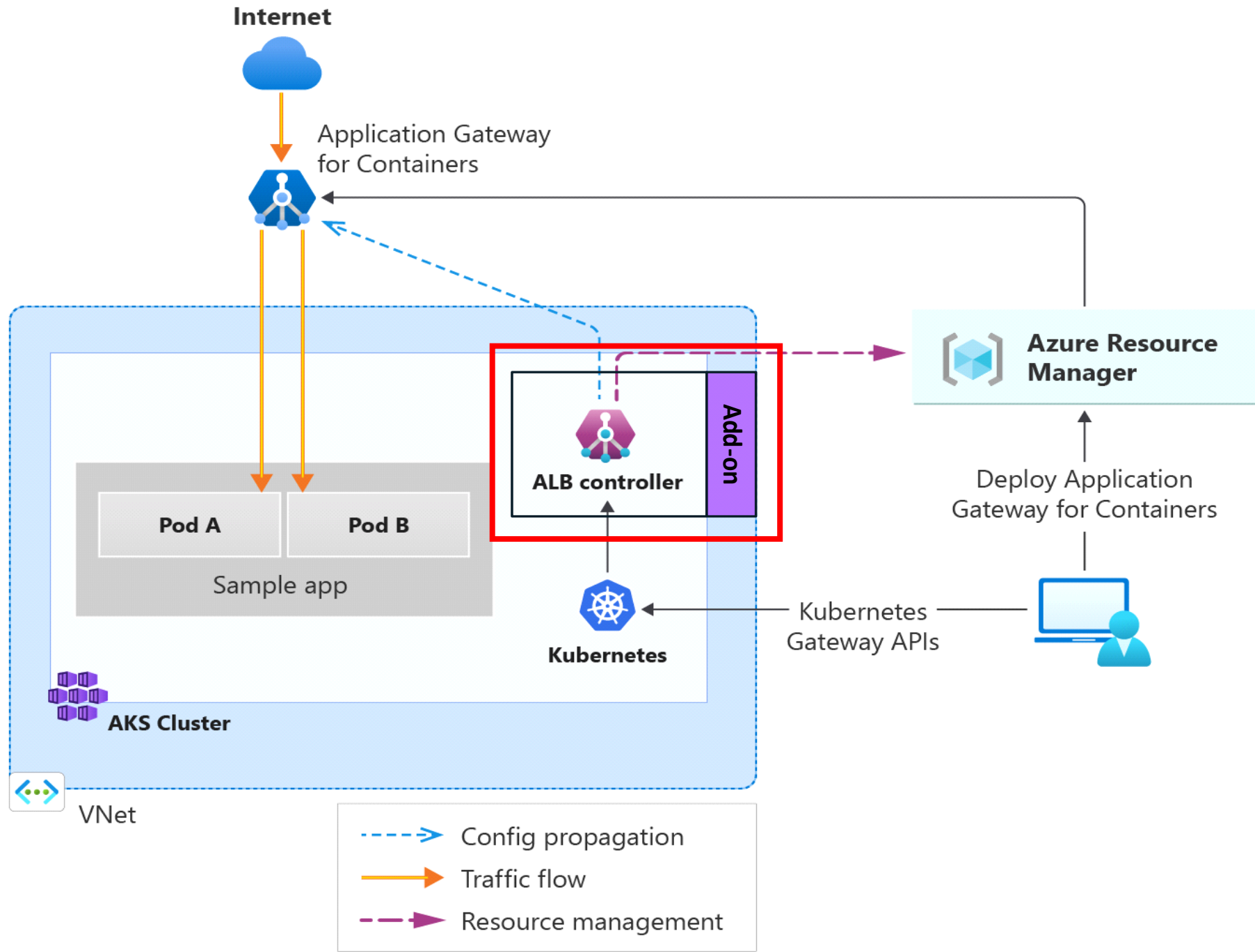
Azure Application Gateway for Containers

Application Gateway for Containers

AKS addon-on in preview

Brings additional features:

- Advanced health checks
- WAF support
- Monitoring
- Autoscaling and high-availability



Demo

Beyond Ingress

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